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Link to All the Completed Modules

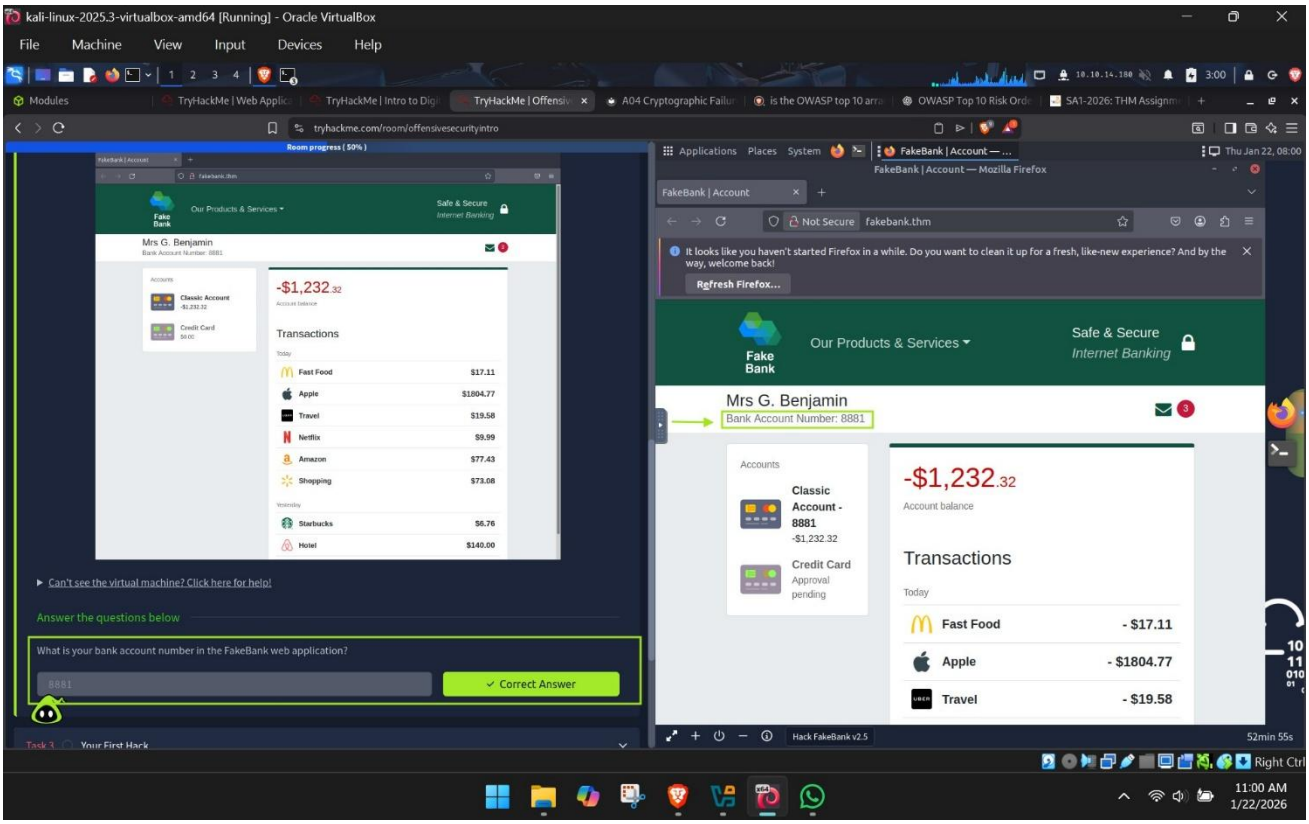
https://tryhackme.com/p/ngugiallan05

Section 1: Offensive Security Intro

Offensive security is all about breaking into computer systems, exploiting software bugs as well as finding loopholes in applications to gain unauthorized access. Its goal is to understand hacker tactics and enhance system defences.

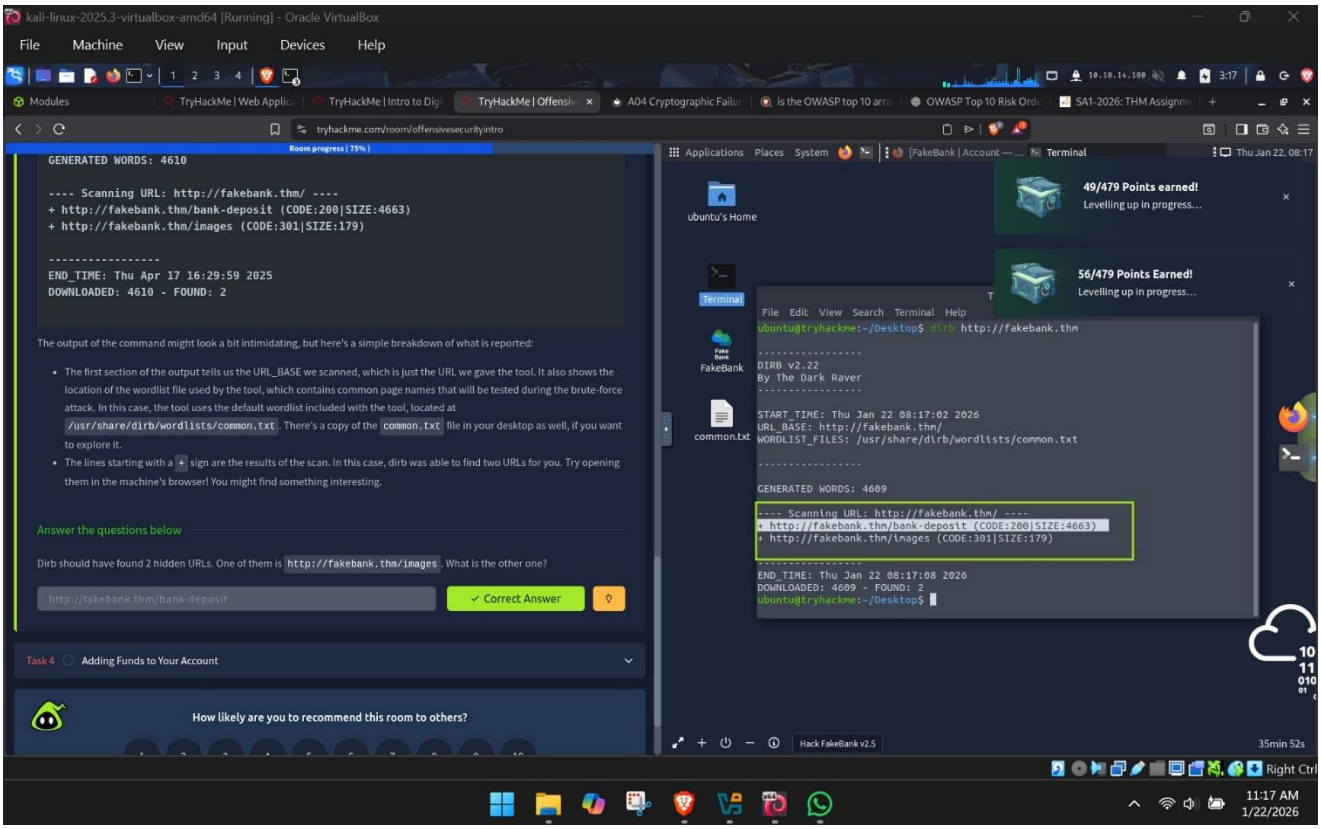
Starting the lab

This task involved starting a fake bank application which was safe to hack. It basically inquired on the bak account number as it appeared on the display.



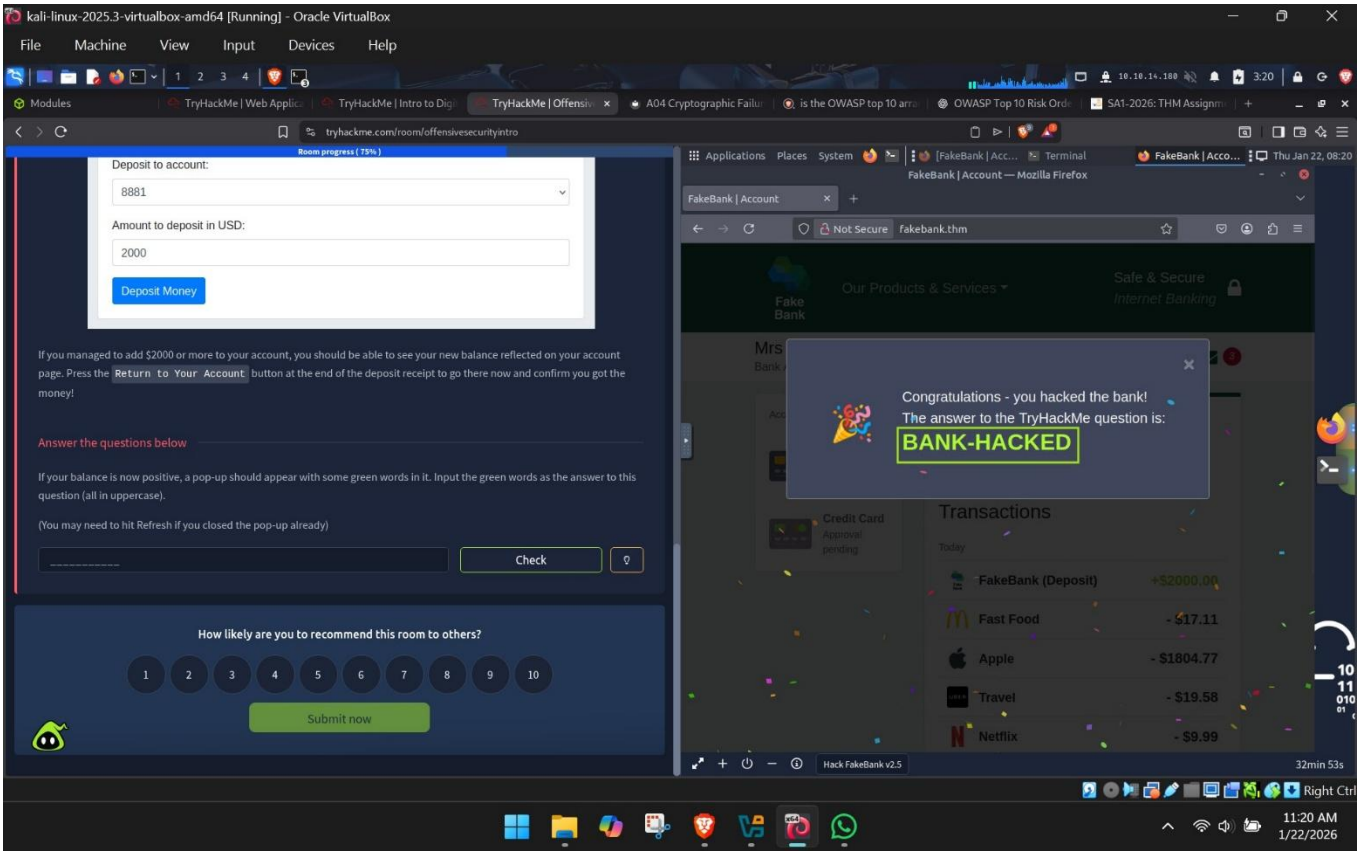
Your First Hack

The goal of this task was to find a way to hack the application in order to steal money. Dirb was used to brute force by taking a list of potential page names and going through them one by one to find if they exist in that application or even a website. The results are shown below:

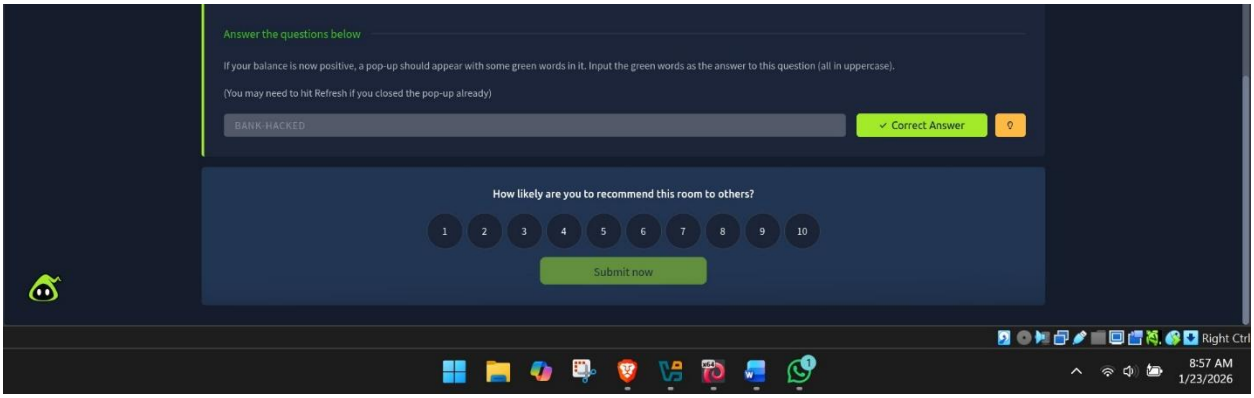


Adding Funds to Your Account

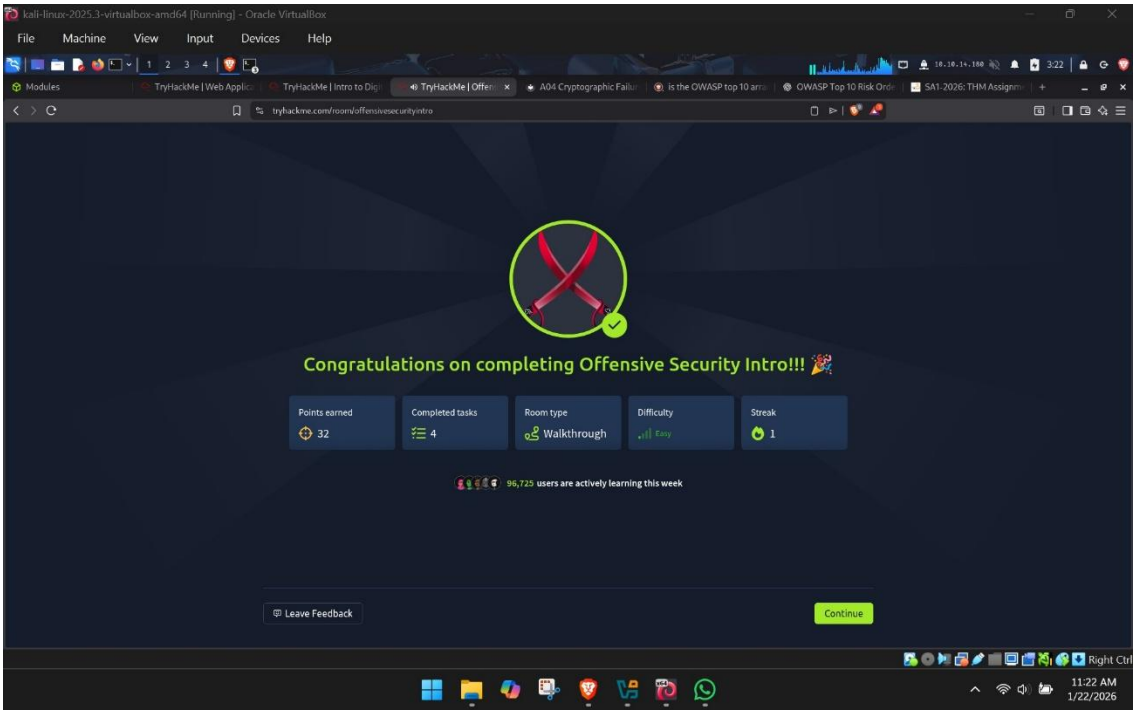
This task involved accessing a secret deposit page from the address bar of the website, and depositing money into the user's account. This is shown below:



Once I got the answer, I input it in the blank space.



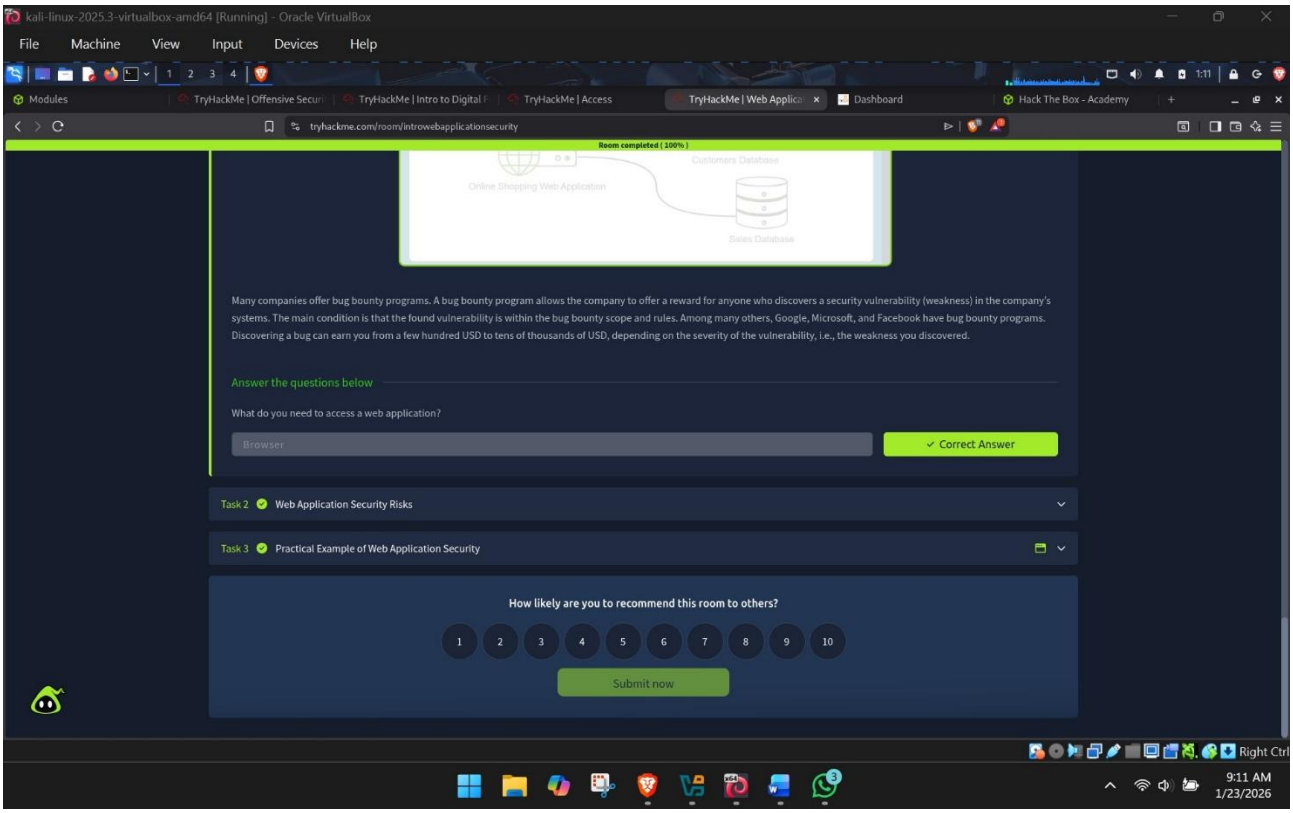
Module Completion



Section 2: Web Application Security

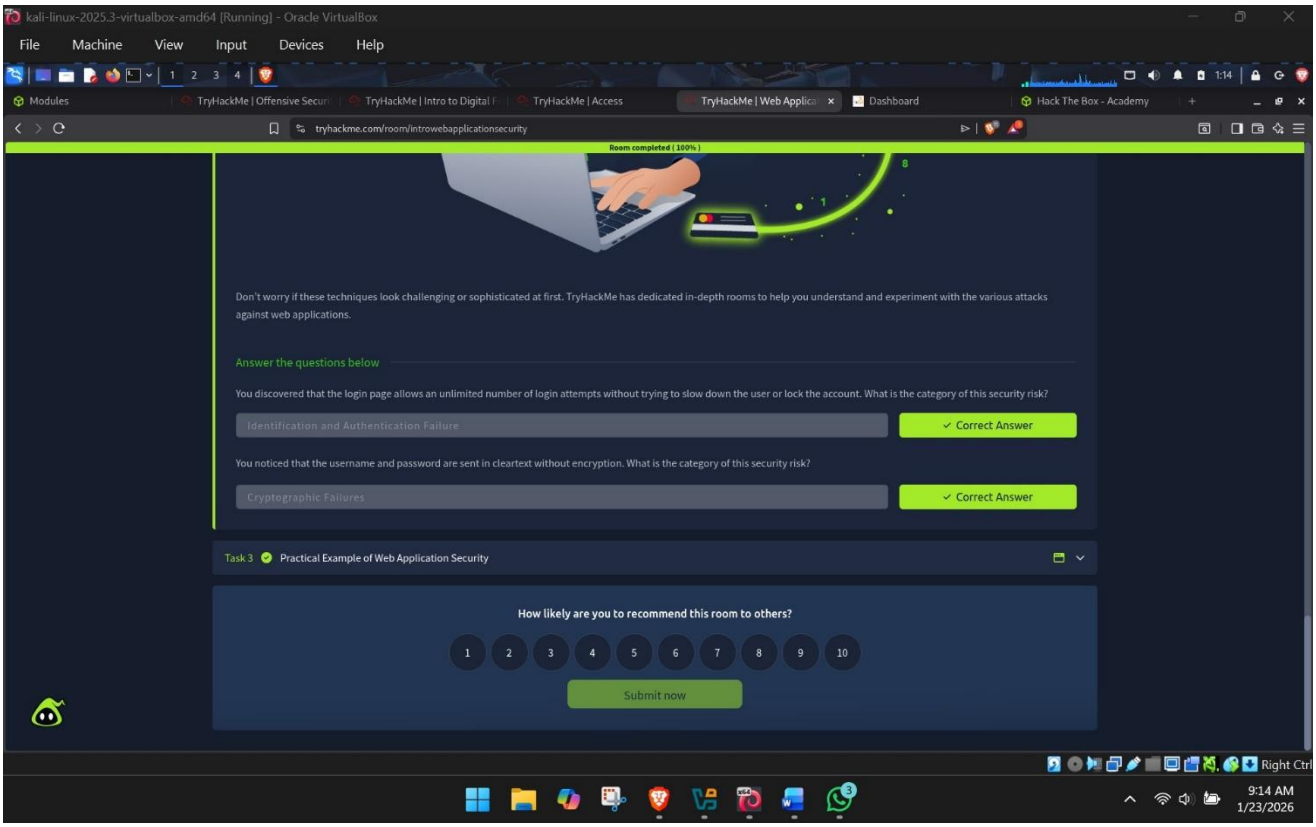
Introduction

In this module, I learned that a web application is an application that is directly utilized once an internet connection is established. By simply navigating to a certain address through the internet, users have an opportunity to use numerous services such as email services, office suits amongst others.



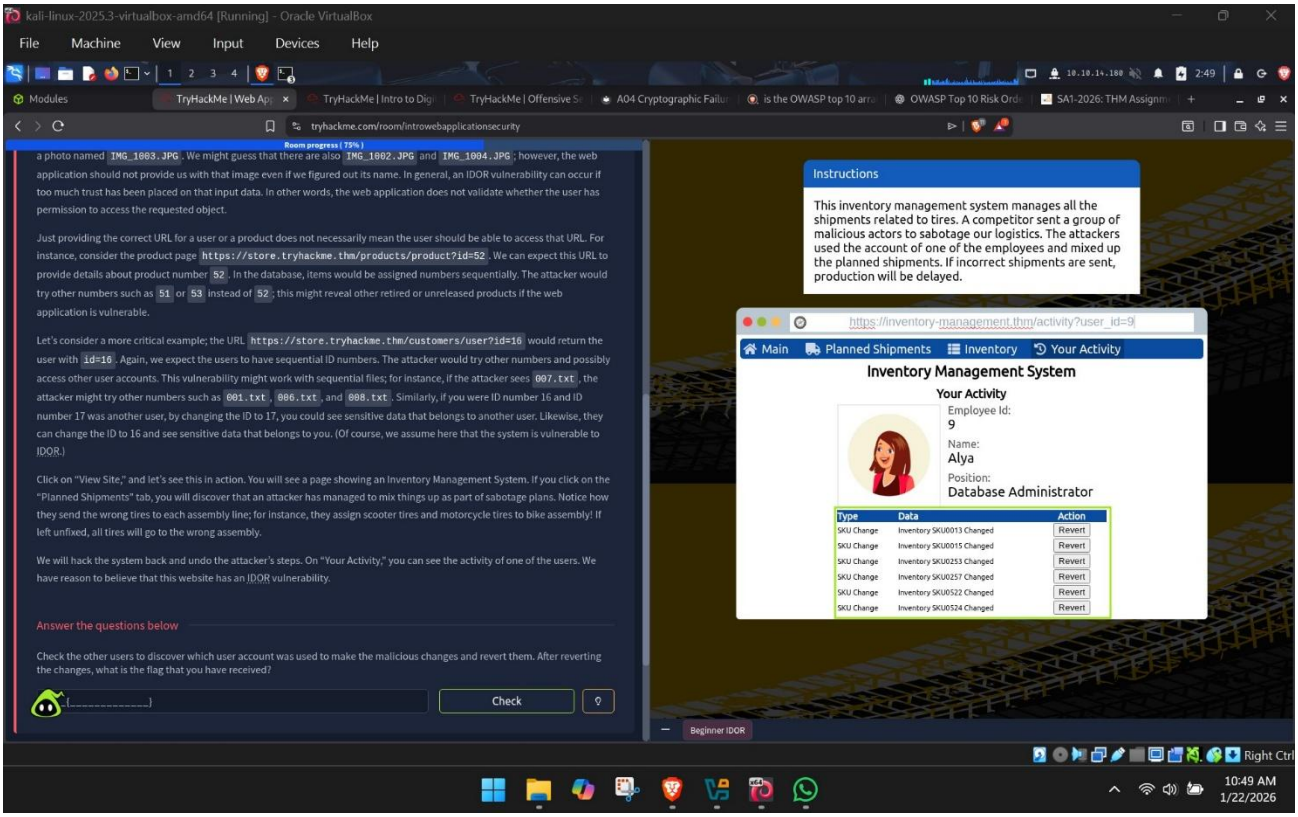
Web Application Security Risks

This task introduced us to the OWASP Top 10, with an emphasis on Identificationa and Authentication, Broken Access Control, Injection and Cryptographic failures, since these are predominant in web applications.

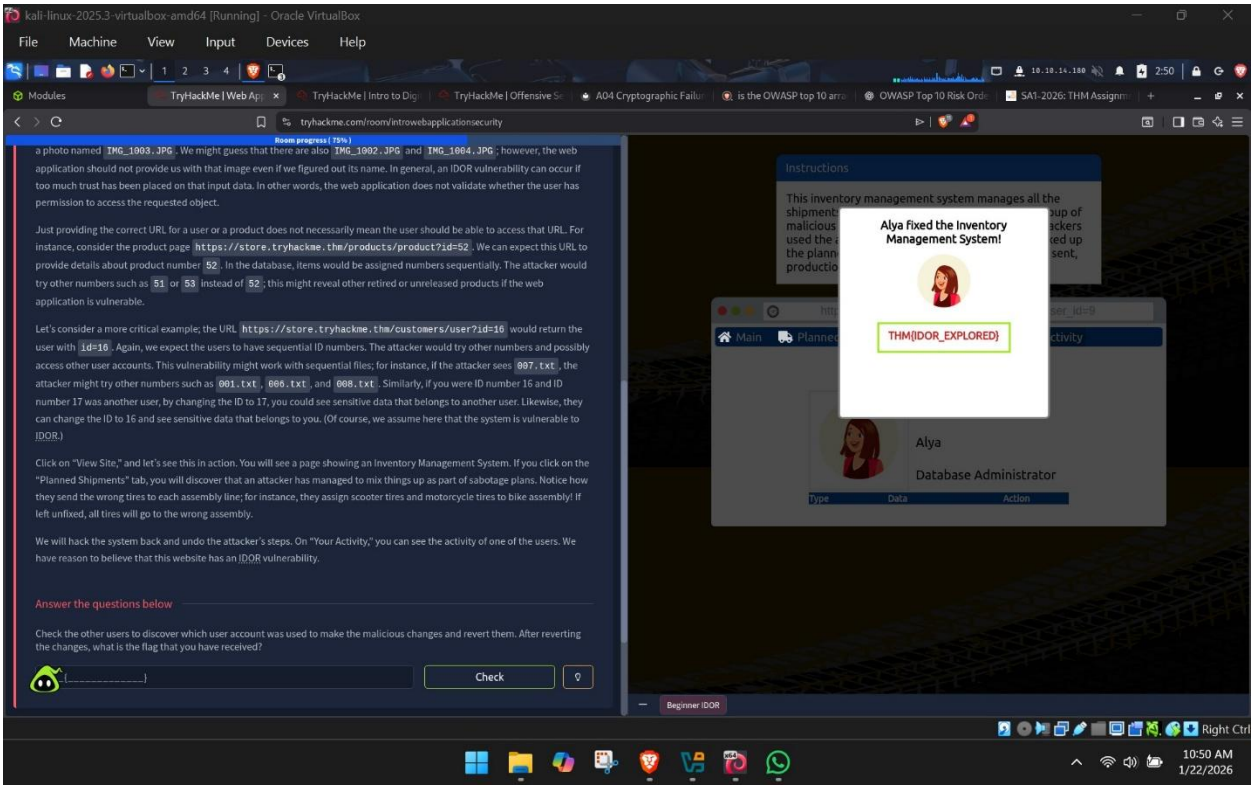


Practical Example of Web Application Security

This task introduced us to a vulnerable web application with an Insecure Direct Object Reference (IDOR) which is a part of Broken Access Control. It involved checking the users within the application to see the malicious changes that were implemented and revert them. This is shown below:

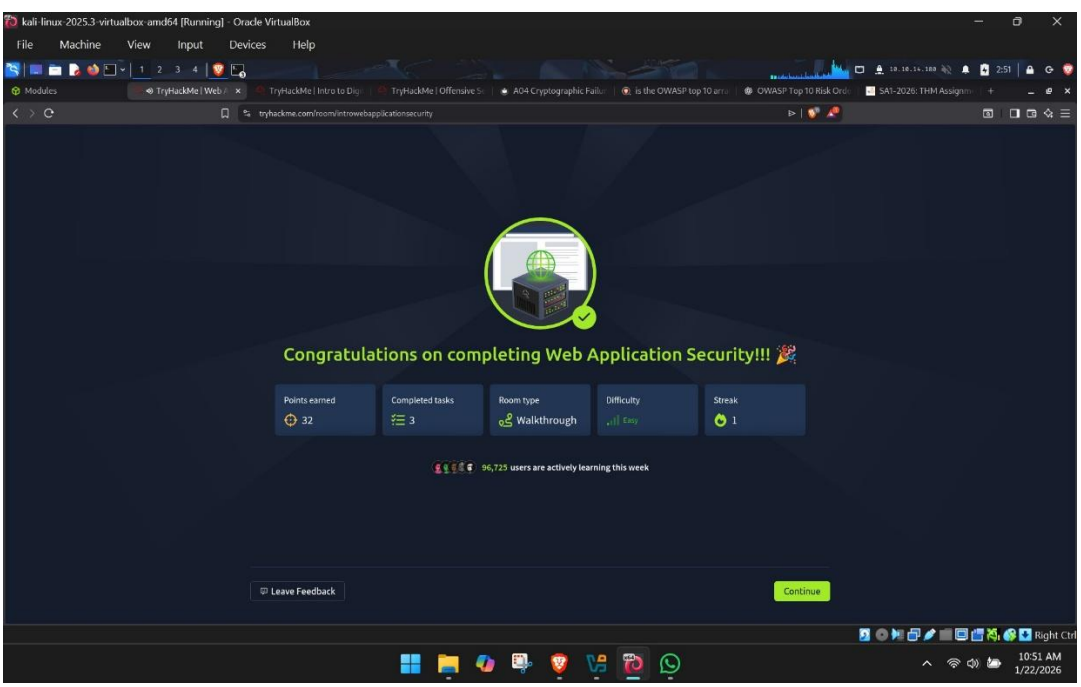


Upon checking the users through the web address, the malicious changes were found through the database administrator, Alya, and was able to revert them.



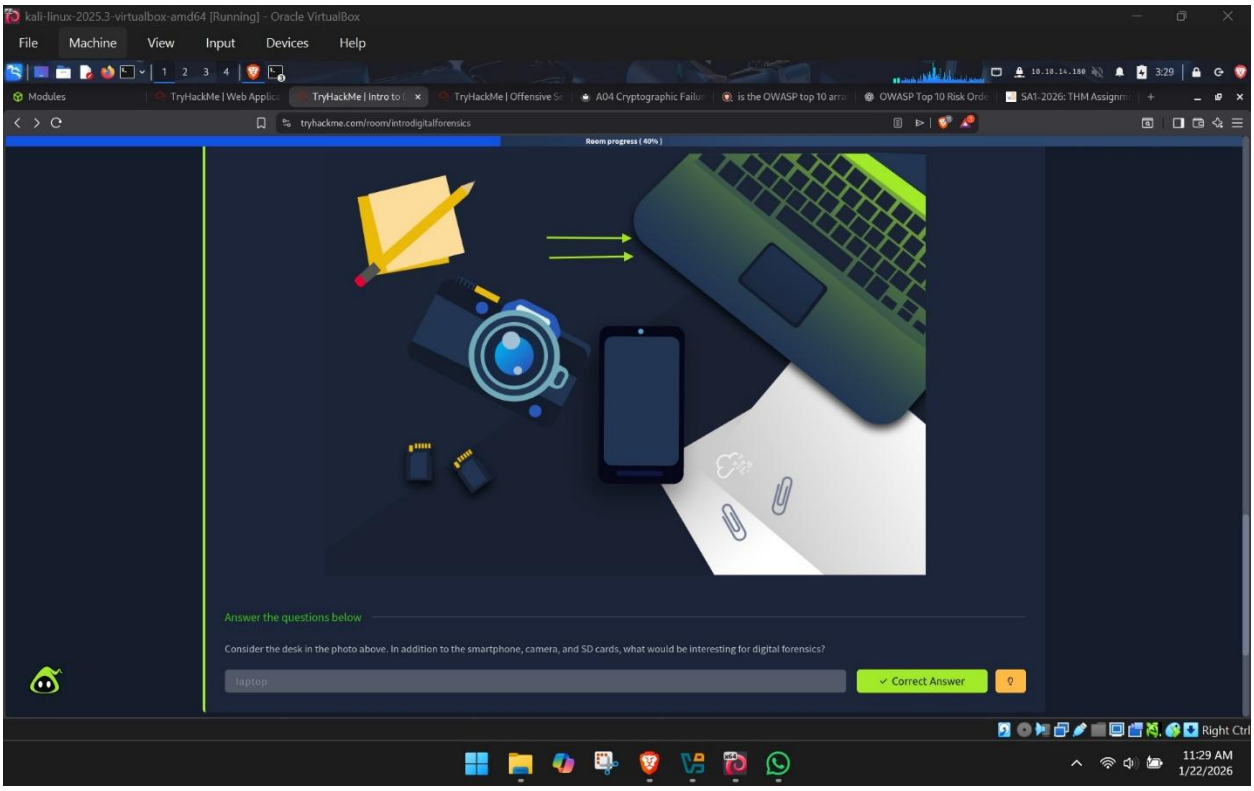
The flag obtained was then used to answer the final question in the task.

Module Completion



Section 3: Introduction to Digital Forensics

Digital forensic science is the use of computer science techniques for the identification, preservation, and investigation of evidence from digital gadgets in the course of legal or corporate investigations. It is the fundamental methodology for the use, evaluation, protection, and inspection of physical evidence, including smartphones, laptops, for enforcing law in the public domain as well as policy breaches in the private domain. By using special techniques, investigators can transform the available huge amount of information into facts.



Digital Forensics Process

This task involves the basics steps a digital forensics investigator should take in a scene with digital evidence, after acquiring legal authorization. It also highlights the process that he or she should take once the evidence is acquired and taken to the lab.

kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

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tryhackme.com/room/introdigitalforensics

Room progress (40%)

3. Return the digital evidence to the secure container: You will be working on the copy. If you damage the copy, you can always create a new one.
4. Start processing the copy on your forensics workstation.

The above steps have been adapted from Guide to Computer Forensics and Investigations, 6th Edition.

More generally, according to the former director of the Defense Computer Forensics Laboratory, Ken Zatyko, digital forensics includes:

- Proper search authority: Investigators cannot commence without the proper legal authority.
- Chain of custody: This is necessary to keep track of who was holding the evidence at any time.
- Validation with mathematics: Using a special kind of mathematical function, called a hash function, we can confirm that a file has not been modified.
- Use of validated tools: The tools used in digital forensics should be validated to ensure that they work correctly. For example, if you are creating an image of a disk, you want to ensure that the forensic image is identical to the data on the disk.
- Repeatability: The findings of digital forensics can be reproduced as long as the proper skills and tools are available.
- Reporting: The digital forensics investigation is concluded with a report that shows the evidence related to the case that was discovered.

Answer the questions below

It is essential to keep track of who is handling it at any point in time to ensure that evidence is admissible in the court of law. What is the name of the documentation that would help establish that?

Chain of Custody

✓ Correct Answer

Task 3 Practical Example of Digital Forensics

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

Submit now

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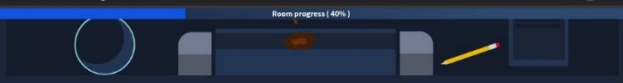
tryhackme.com/room/introdigitalforensics

Room progress (40%)



As a digital forensics investigator, you arrive at a scene similar to the one shown in the image above. What should you do as a digital forensics investigator? After getting the proper legal authorization, the basic plan goes as follows:

1. Acquire the evidence: Collect the digital devices such as laptops, storage devices, and digital cameras. (Note that laptops and computers require special handling if they are turned on; however, this is outside the scope of this room.)
2. Establish a chain of custody: Fill out the related form appropriately (Sample form). The purpose is to ensure that only the authorized investigators had access to the evidence and no one could have tampered with it.
3. Place the evidence in a secure container: You want to ensure that the evidence does not get damaged. In the case of smartphones, you want to ensure that they cannot access the network, so they don't get wiped remotely.
4. Transport the evidence to your digital forensics lab.

At the lab, the process goes as follows:

1. Retrieve the digital evidence from the secure container.
2. Create a forensic copy of the evidence: The forensic copy requires advanced software to avoid modifying the original data.
3. Return the digital evidence to the secure container: You will be working on the copy. If you damage the copy, you can always create a new one.
4. Start processing the copy on your forensics workstation.

The above steps have been adapted from Guide to Computer Forensics and Investigations, 6th Edition.

More generally, according to the former director of the Defense Computer Forensics Laboratory, Ken Zatyko, digital forensics includes:

- Proper search authority: Investigators cannot commence without the proper legal authority.
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- Reporting: The digital forensics investigation is concluded with a report that shows the evidence related to the case that was discovered.

Answer the questions below

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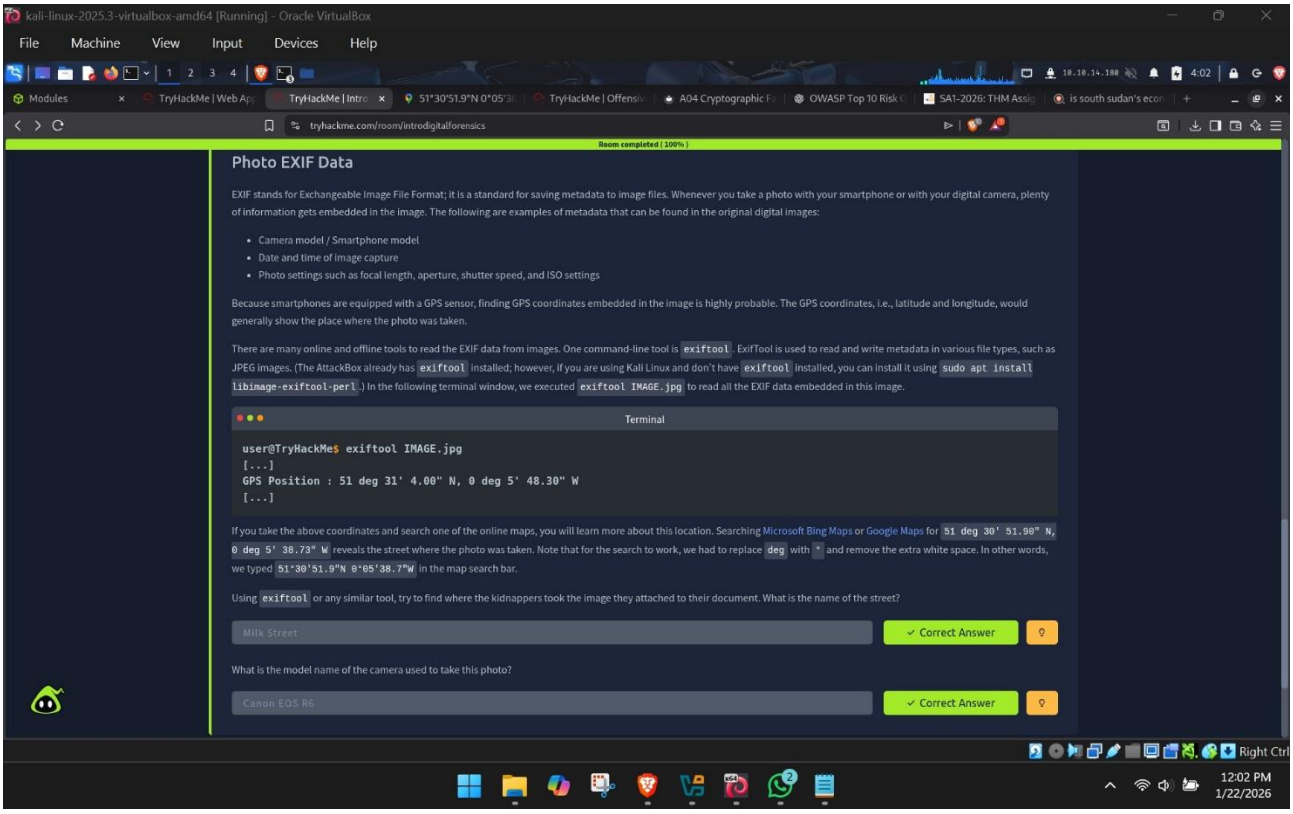
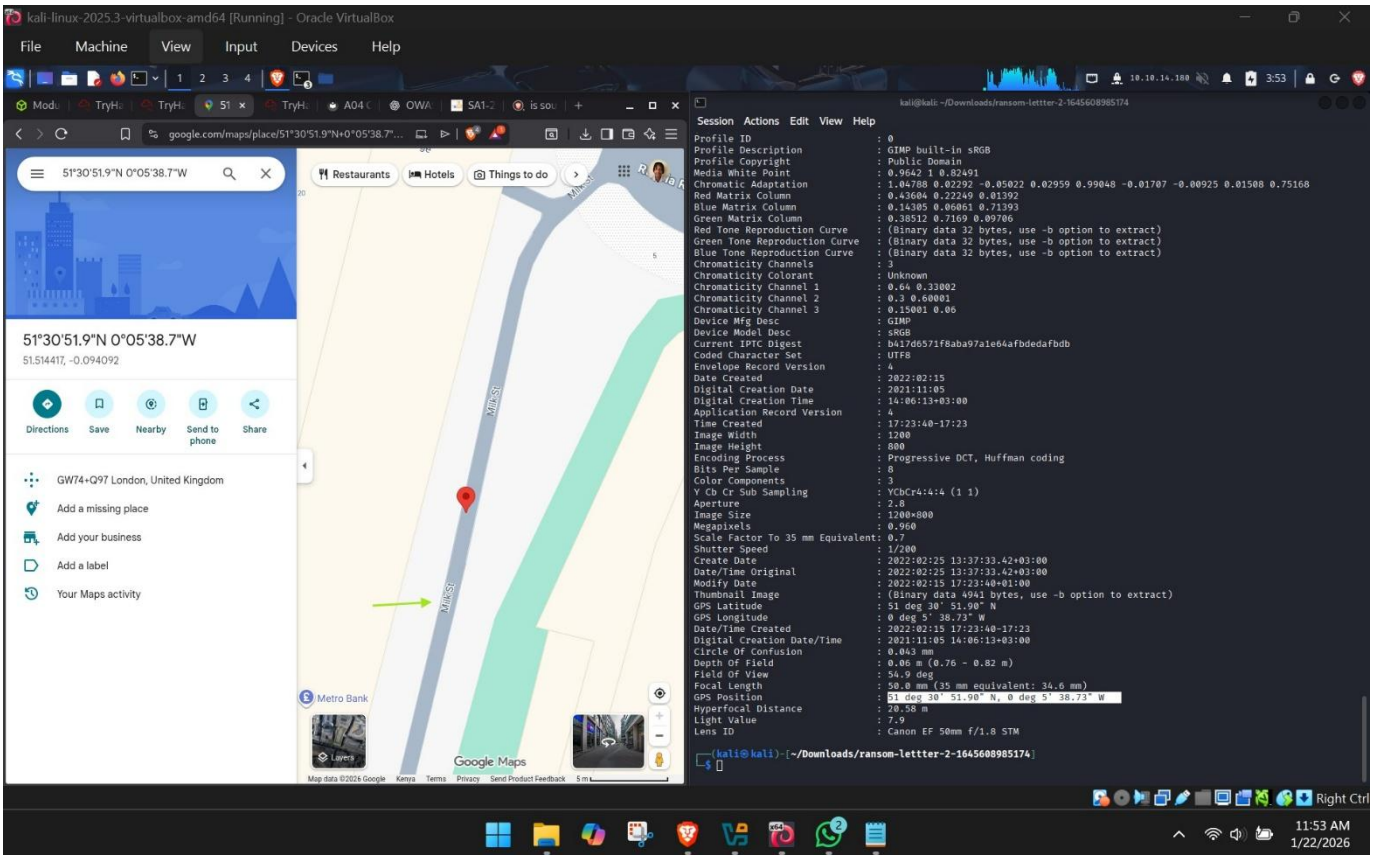
The chain of custody is a document that documents the persons that came into contact with digital evidence, ensuring that no one tampered or interfered with it.

Practical Examples of Digital Forensics

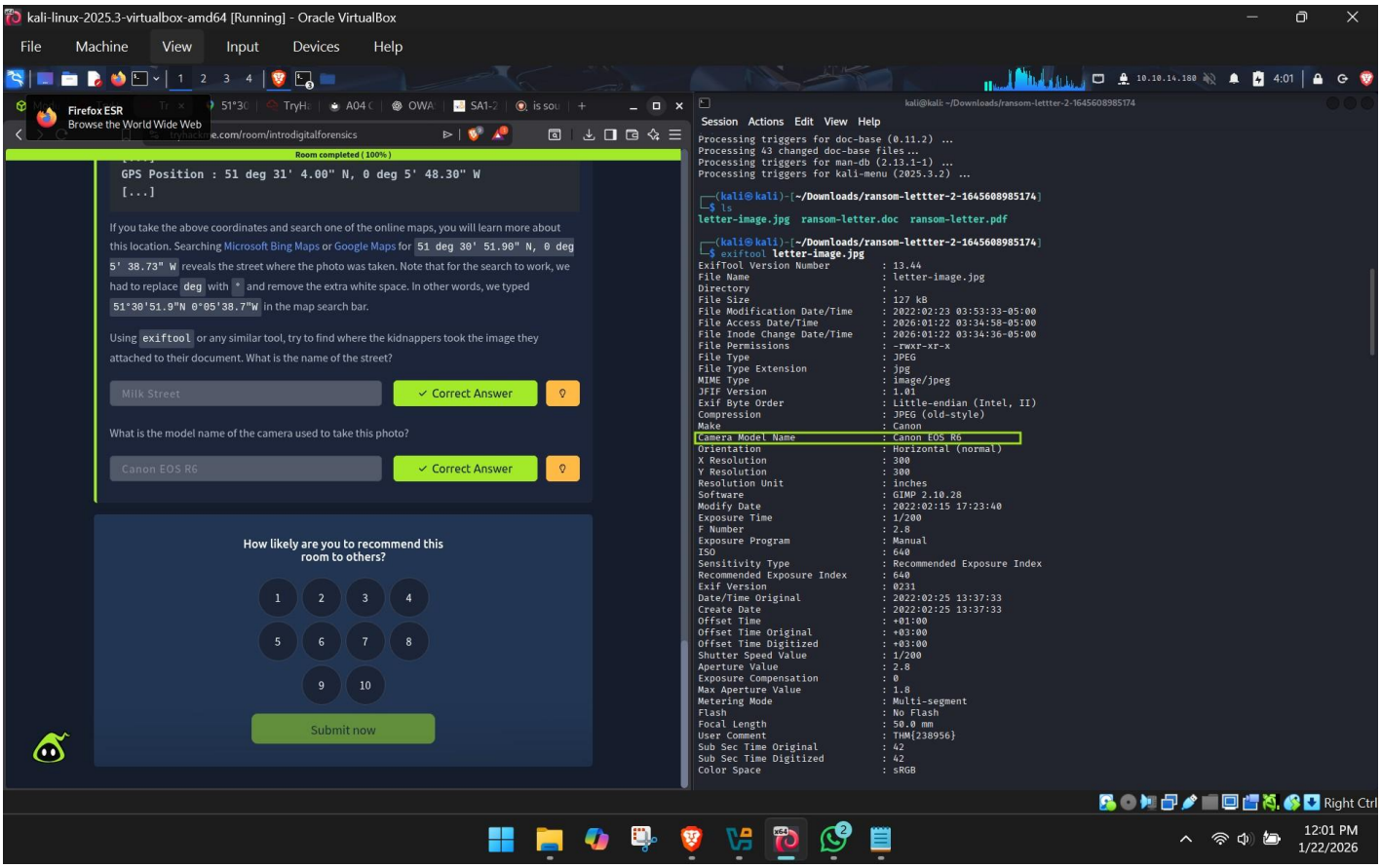
This task involved applying the knowledge obtained through the previous tasks, and applying exiftool that enables one view the metadata of a digital piece of evidence like an image or a word or pdf file. The process is shown below.



The questions included in this task involved using a tool (exiftool) to retrieve metadata from an image file. The first question required one to find the location where they took the image attached to the documents, which was derived in Milk Street as shown below:



The second question involved finding the model name of the camera used to take the image, which is still within the metadata retrieved. The model name was found in the image below:



Conclusion

The three tasks introduced me to the basics of cybersecurity, with a focus on offensive security, web application security and digital forensics. It was very exciting and fun going through these tasks. These tasks enabled me gain knowledge, and laid a good foundation into what I will learn more in this discipline.